Example 2.2

Find the value of the real numbers x and y, if the complex number $(2+i)x+(1-i)y+2i-3$ and $x+(-1+2i)y+1+i$ are equal

Solution

Let $z_1 = (2+i)x+(1-i)y+2i-3=(2x+y-3)+i(x-y+2)$ and

$z_2 = x+(-1+2i)y+1+i=(x-y+1)+i(2y+1)$.

Given that $z_1 = z_2$.

Therefore $(2x+y-3)+i(x-y+2)=(x-y+1)+i(2y+1)$.

Equating real and imaginary parts separately, gives

$$2x+y-3 = x-y+1 \quad \Rightarrow \quad x+2y=4.$$
$$x-y+2 = 2y+1 \quad \Rightarrow \quad x-3y=-1.$$

Solving the above equations, gives

$$x=2 \text{ and } y=1.$$

EXERCISE 2.2

1. Evaluate the following if $z=5-2i$ and $w=-1+3i$

   (i) $z+w$ (ii) $z-iw$ (iii) $2z+3w$

   (iv) zw (v) $z^2+2zw+w^2$ (vi) $(z+w)^2$.

2. Given the complex number $z=2+3i$, represent the complex numbers in Argand diagram.

   (i) z, iz, and $z+iz$ (ii) z, $-iz$, and $z-iz$.

3. Find the values of the real numbers x and y, if the complex numbers $(3-i)x-(2-i)y+2i+5$ and $2x+(-1+2i)y+3+2i$ are equal.

2.3 Basic Algebraic Properties of Complex Numbers

The properties of addition and multiplication of complex numbers are the same as for real numbers. We list here the basic algebraic properties and verify some of them.

2.3.1 Properties of complex numbers

The complex numbers satisfy the following properties under **addition.**	The complex numbers satisfy the following properties under **multiplication.**
(i) **Closure property** For any two complex numbers z_1 and z_2, the sum z_1+z_2 is also a complex number.	(i) **Closure property** For any two complex numbers z_1 and z_2, the product $z_1 z_2$ is also a complex number.

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